

Ray Doan

+84 869 033 950 | raydoan.contact@gmail.com | linkedin.com/in/iamraydoan | github.com/iamraydoan

EDUCATION

VNU University of Engineering and Technology

Bachelor's in Information Technology, GPA: 3.64/4.0

Hanoi, Vietnam

Aug. 2017 – June 2021

EXPERIENCE

Software Engineer

July 2023 – Present

VTI

Hanoi, Vietnam

- Engineered real-time IoT, MES, and WMS solutions for manufacturing plants, improving production visibility by **35%** and reducing order fulfillment latency by **40%**
- Built high-throughput MQTT telemetry services and live dashboards processing **5M+ daily events** from **200+ factory machines** with **¡200ms latency**
- Architected CDC pipelines streaming data from MongoDB and PostgreSQL into **ClickHouse**, reducing analytical query times from **15s to ¡300ms** on datasets of **10M+ records**
- Developed automated streaming report generation pipelines supporting exports of up to **1M rows** (Excel max limit) without memory overflow

Software Engineer

Aug. 2022 – June 2023

EVNFinance

Hanoi, Vietnam

- Architected backend microservices (NestJS, PostgreSQL, MongoDB) for lending solutions, processing **100,000+ loan applications** with **99.9% uptime**
- Developed backend for an investment mobile app using **Kafka** and **Docker**, handling **50,000+ active daily users** during peak trading hours
- Optimized database queries and implemented Redis caching, reducing overall API response latency by **45%**

Software Engineer

June 2021 – July 2022

Merchize

Hanoi, Vietnam

- Developed real-time 2D/3D product mockup rendering engines using **Node.js**, **Canvas**, and **Three.js**, reducing render times by **60%** (from **3s to ¡1.2s**) and boosting conversion by **15%**
- Integrated Three.js and React to enable interactive 3D product views, enhancing user experience for **10,000+ active sellers**
- Built tools for adding and adjusting designs across product templates, collaborating with the design team on usability

Software Engineer Intern

June 2018 – May 2021

Merchize

Hanoi, Vietnam

- Built multi-platform e-commerce integration services (Amazon, eBay, Etsy) synchronizing **20,000+ daily orders** and real-time shipment status
- Developed ad-spy monitoring tools and Chrome extensions tracking **5M+ ad creatives** across **100K+ campaigns** on Amazon, Etsy, and Facebook
- Implemented Chrome extensions for real-time ad monitoring and campaign optimization

PROJECTS

IoT & MES Real-Time Analytics Pipeline | *NestJS, MQTT, ClickHouse, PostgreSQL, Redis* Jan. 2024 – Present

- Engineered high-throughput IoT telemetry pipeline processing **5M+ daily events** from 200+ machines with **¡200ms latency**
- Architected CDC streaming from PostgreSQL/MongoDB to ClickHouse, cutting analytical query response times from **15s to ¡300ms** on **10M+ rows**
- Developed automated streaming report generator exporting up to **1M rows** per file (Excel max limit) without memory issues

3D Product Customizer & Order Sync Engine | *Node.js, React, Three.js, Python, MongoDB, Docker* June 2021 – May 2022

- Implemented dynamic 3D mockup rendering system with Three.js & Canvas, reducing render time by **60%** (**¡1.2s** load time)
- Built multi-platform order sync service integration connecting Amazon, eBay, and Etsy stores handling **20,000+ daily orders**
- Created automated Chrome extensions for real-time ad monitoring, tracking **5M+ ad creatives** across **100K+ campaigns**

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, SQL

Frameworks & Libraries: Node.js, NestJS, ReactJS, React Native, Redux, Three.js, Tailwind CSS

Databases & Messaging: MongoDB, PostgreSQL, MySQL, ClickHouse, Redis, Kafka, RabbitMQ, NATS, MQTT

Architecture & Tools: Microservices, Event-Driven Architecture, Docker, CI/CD, Git, Linux